PROJECT 9

FUEL CONSUMPTION

Regression Assumptions for the Fuel Consumption Model

PRESENTED BY MUNAWAR

TOPIC REVIEW - ASSUMPTIONS

Regression model assumptions are rules or requirements that must be met for regression analysis results to be accurate and reliable. Regression is used to examine the relationship or influence of one variable on another.

Assumptions in a regression model are conditions that must be met by the data so that the linear regression model provides accurate, valid, and unbiased estimation results. Regression Model has several assumptions:

- Validity
- Representativeness
- Additivity and Linearity
- Independence from errors
- Equal variance of errors
- Normality of errors

BACKGROUND

- In this case, we analyze the relationship between *Hwy* (fuel consumption on the highway, in L/100 km) and *Fuel_Consumption* (combined fuel consumption, in L/100 km) — two key measures of a vehicle's operational efficiency.
- A lower *Hwy* value indicates better fuel economy on the highway, meaning the vehicle consumes fewer litres per 100 km under highway driving conditions, which generally reflects a vehicle's aerodynamic qualities and transmission efficiency.
- The purpose of this analysis is to compare three regression modeling approaches to find the most accurate model in capturing the correlation between highway fuel consumption (Hwy) and total combined fuel consumption (*Fuel_Consumption*).

OBJECTIVE

Compare three regression modeling approaches to find the most accurate model for predicting combined fuel consumption from highway fuel consumption.

- Linear Regression
- Log Transform Model
- Reciprocal Transform

MODEL 1 - LINEAR REGRESSION

$$fuel\_\text{consumption} = -0.138 + 1.340Hwy$$

Key Takeaway

- For every 1 L/100 km increase in Hwy, combined FC increases by 1.34 L/100 km on average.
- 2 L/100 km rise in Hwy predicts a 2.68 L/100 km rise in combined fuel consumption.
- R^2 = 0.8079 — the model explains 80.79% of the variance in combined FC.
- Residual histogram approaches a bell curve — normality assumption largely met.

⚠Limitation: Assumes rigid linearity; predictions may be inaccurate outside the observed Hwy range; doesn't account for vehicle-specific factors.

MODEL 2 - LOG TRANSFORM REGRESSION

$$fuel\_\text{consumption} = \alpha + \beta \times log(Hwy)$$

Key Takeaway

- Log transformation compresses the scale of Hwy, reducing the impact of extreme values.
- Better captures diminishing returns — vehicles with very high Hwy values show less additional combined FC increase.
- R^2 = 0.8118 — improves upon the linear model by explaining 81.18% of variance.
- Interpretation changes: a 1-unit increase in log(Hwy) corresponds to a percentage-based change in Hwy.

MODEL 3 – RECIPROCAL TRANSFORM REGRESSION

$$fuel\_\text{consumption} = \alpha + \beta \times (1/Hwy)$$

Key Takeaway

- The reciprocal transformation inverts the Hwy scale — vehicles with very low Hwy (most efficient) get the highest 1/Hwy values.
- R^2 = 0.773 — the lowest of the three models, explaining only 77.30% of the variance.
- Performs worse than both the linear and log models, suggesting the reciprocal functional form does not suit this data well.
- This model is not recommended for predicting combined fuel consumption from highway FC.

MODEL COMPARISON

Model	R^2
Linear	80.79%
Log Transform	81.18%
Reciprocal Transform	77.30%

Log Transform Model achieves the highest R^2 with 81.18%, providing the best predictive accuracy.

CONCLUSION

1

Best Model

Log Transform (log Hwy) achieves R^2 = 81.18%, the highest among all three models tested.

2

Baseline Model

Linear model is strong (R^2 = 80.79%) and easier to interpret, suitable as a quick baseline.

3

Avoid Reciprocal

Reciprocal transform (R^2 = 77.3%) underperforms and is not recommended for this dataset.

4

Recommendation: While the current log model already provides high explanatory power, further improvements can be explored by adding predictors such as engine size, vehicle class, or transmission type to build a more comprehensive combined fuel consumption model.

THANK YOU FOR YOUR TIME

github.com/nawar27

munawarrf8@gmail.com

DATA ANALYST PRESENTATION